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Software Reliability Estimation Method Based on Markov Usage Models Using Importance Sampling

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Abstract— **Importance sampling (IS) is a change-of-measure technique for speeding up the simulation of rare events in stochastic systems. In this paper, we presented an approach uses Importance Sampling technique for efficient estimation of software reliability via Markov software usage models in statistical testing. By suitable changes of the probabilities of state transitions during test, an iterative method based on the Ali-Silvey distance is proposed for this choice, and an unbiased reliability estimator with zero variance is obtained. A learning algorithm for the computation of optimal transition probabilities of the Markov chain usage model is also presented and experimental results of this algorithm are reported.**

I. Introduction

SOFTWARE plays an increasingly important role in today's society. Reliable software is a prerequisite for most functions in our daily life. Software reliability can be viewed as a good measure of quantifying software quality and is defined as the probability of failure-free software operation for a specified period of time in a specified environment [1]. A very common approach is to simply use it as a synonym for correctness. The current status of software reliability assessments is far from good [2].

The software reliability depends not only on the defects residing in the software systems, but also on how the software is used, i.e. the usage profile [3]. The usage profile, or the operational profile, is the characterization of the users' operative utilization of a software system. The usage is characterized in terms of the user initiated events and the probability for these events. A Markov usage model is based on a Markov chain which characterizes the distribution of the possible uses of a given program in its intended environment [4]-[8]. Such a model is required whenever unbiased estimates of quantities like reliability or risk are to be derived from the results of tests. The statistical testing method allowing such estimate is random testing with suitably chosen input distributions, and the most direct way to achieve the goal of unbiased estimation consists in choosing the operational profile. During statistical testing, the test cases are chosen according to the operational use distribution and applied to the software under test, such that the frequency of failures during the test leads to unbiased estimates of the probability of failures in the application field.

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A serious drawback of this technique has been outlined by Walton et al. [4]: The most frequently used operations are always given the most testing irrespectively of another important criterion, i.e. the relative error-proneness of the single operations. Intuitively, one would like to provide more test effort for operations that are expected to be especially error-prone, even if they are not very frequently used. They proposed three solutions to this problem, however their application implies that the relative failure frequency during test does not yield an unbiased reliability estimate anymore. Gutjahr [14]-[17] discussed an approach to overcome this problem by replace the original empirical usage profile with a special test profile. In this approach, transition probabilities during testing are optimized with respect to a specific dependability measure taking into account critical or error-prone operations. As a result of this optimization unbiased dependability measures can be obtained. Other approaches where critical sections of the program code are analyzed do not provide the tester with appropriate reliability estimates.

Let us emphasize that the drawback of conventional Markov usage models mentioned by Walton et al is especially prohibitive in the case of safety-critical software, where there are always operations with low use frequency and high criticality. So, without a technique overcoming this drawback, the applicability of Markov usage models to this type of software would be very limited.

II. Relevant Previous Work

Interesting work on the generation of test cases has been carried out at the University of Tennessee at Knoxville. Using a Markovian assumption, Whittaker [6] and Walton [7] have developed both theoretical frameworks and practical tools for generating testing trajectories. Whittaker and Poore [8] have proposed using Markov chains for modeling sequences of inputs to a software system, described usage for the purpose of generating testing cases and to guide software testing statistically analysis. Doerner an Gutjahr [9] have presented a technique for obtaining appropriate software test sequences from a Markov usage model description of the expected use of a software product. Their approach addresses the tradeoff between testing costs and failure risk caused by untested processing steps. Wohilin and Runeson [10] also use Markov chains for usage modeling, specifically for the reliability engineering of software components. Srivastava et al. [11] have proposed a technique to generate optimized test sequences from a Markov chain based usage model.

The proposed technique uses an colony optimization as its basis and also incorporates factors like cost and criticality of various states in the model.

In several papers, Woit [5] has discussed the problem of estimating reliability of a module after a long series of successful tests. Woit introduces the idea of using a set of conditional probabilities (an operational profile) to generate testing trajectories whose statistical characteristics are those that can be expected when the module is actually used. After testing the module with these traces, statistical methods can be used to estimate the reliability of the module if it is used in a way consistent with the operational profile. Thomason and Whittaker [12] have given the Poisson and exponential distribution as approximations in software reliability analysis by treating software failure as a rare event with a finite-state, discrete parameter, recurrent Markov chain. Prowell and Poore[13] have proposed an analytical computation for the reliability of a system, the reliability estimator takes into account individual variation among tests, can make use of prior testing information, and presented a Bayesian reliability model that is gaining support in field use. Gutjahr [14] use the Importance Sampling technique from Rare Event Simulation for the software failure cost estimation of Markov usage model. He has shown that by a shift the transition probability of the Markov chain, obtained an unbiased failure cost estimator with reduced variance. Furthermore, it is extended to the software (un)reliability estimation [15], failure risk estimation [16] and software dependability evaluation [17]. Similar problems have already been investigated by Yan et al. [18],[19], Zhang et al.[20], [21] and Xu et al. [22].

The aim of the present paper is to develop a method of sampling test cases from a Markov software usage model based on Importance Sampling method [14]-[22] , such that

1) test results can still be used to compute unbiased estimates of reliability.
2) prior knowledge on the error-proneness of each operation can be used for a zero-variance of the reliability estimator.

The idea is to expose the transition probabilities in the Markov chain to a probability shift, as suggested by Walton et al. [4], but to compensate this shift afterwards by assigning weights to the test results. Essentially, this is the importance sampling technique of Rare Event Simulation. Considering prior estimates of the failure probabilities connected with the operations (arcs) in the Markov chain are given, the optimal transition probability shift results as the solution of a certain stochastic optimization problem.

The paper is outlined as follows: in Section 3, we formulate the problem in a Markov usage model. Next, we presents a learning algorithm based on the Importance Sampling method to optimize optimal transition probability shift in Section 4. Then, we presents simulation results for software reliability estimation in Section 5. Finally, we give the conclusions in Section 6.

III. Markov Software Usage Models

A Markov usage model can be described by a Markov chain with a unique initial state, a unique final state, and other states. As input we take the Markov chain represented by a strongly connected directed graph $G = (V, A)$ without multiple arcs, and a function $p : V \times V \to [0, 1]$ with following properties:

1) $V = \{1, 2, \cdots, n\}$ is a set of nodes, representing the program states.
2) A is a set of arcs, representing state transitions which always correspond to specific operations of the program. An arc from state i to state j is denoted by the ordered pair (i, j).
3) $p(i, j)$ is the transition probability from state i to state j, if (i, j) is an arc. Otherwise, we set $p(i, j) = 0$. The transition probabilities satisfy the condition $0 \leq p(i, j) \leq 1$ and

$$\sum_{j=1}^{n} p(i, j) = 1, i = 1, 2, \cdots, n.$$

It is always assumed that state 1 is the initial state and state n is the final state. state n is assumed to be an absorbing state, i.e., it cannot be left anymore: $p(n, n) = 1$ and $p(n, j) = 0$ for all $j \neq n$. Furthermore, we assume that each node $i \in V$ is reachable from node 1, i.e., there is a directed path in G from node 1 to node i.

For generating a statistical test case based on this model, we have to step through the chain according to the transition probabilities. The sequence of visited states becomes the test case. Let $\mathbf{X} = (X_1, X_2, \cdots)$ denote the test case of a program π, and P denote the operational profile (or operational distribution) of π, i.e., the probability of distribution representing the use of π in real applications. We set

$$\mathbf{I}_f(\mathbf{x}) = \begin{cases} 0, & \text{if input } \mathbf{x} \text{ is processed correctly by } \pi, \\ 1, & \text{otherwise.} \end{cases}$$

Therein, the decision on correctness is based on the program specification for π, which is considered as given and fixed. By means of the quantities $\mathbf{I}_f(\mathbf{x})$, the reliability of a program π is given by

$$R(\pi) = 1 - \mathbb{E}_P[\mathbf{I}_f(\mathbf{X})] \triangleq 1 - \ell, \tag{1}$$

where $\mathbf{X}$ is a random input according to operational distribution P, and $\mathbb{E}_P$ is the expectation with respect to the operational distribution P, ℓ denotes the unreliability of the program π, i.e., $\mathbb{E}_P[\mathbf{I}_f(\mathbf{X})]$. Since $\mathbf{X}$ is unknown before testing, $R(\pi)$ is also unknown before testing. By means of the test results, it is possible to estimate $R(\pi)$.

IV. Reliability Estimation with Importance Sampling Method

A. *Probability Changes of Reliability Estimation*

Our approach based on the importance sampling technique of Rare Event Simulation, which prescribes that a probability change in a simulation has to be compensated by adjusting suitable weights to the outcomes of the simulation run.

A straightforward unbiased estimator for the reliability of a program π is

$$\widehat{R}(\pi) = 1 - \widehat{\ell} = \frac{1}{N}\sum_{i=1}^{N} I_f(\mathbf{x}_i), \tag{2}$$

where N test cases $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N$ according to the operational distribution P. The total probability of the test case $\mathbf{x}$ is a function of the operational distribution P:

$$f(\mathbf{x}, P) = \mathbb{P}(\mathbf{X} = \mathbf{x}) = \prod_i p(x_i, x_{i+1}). \tag{3}$$

However, the precision of this estimate is still low: Observe the software failures can be considered as rare events, so in most cases reliability $R(\pi)$ will be overestimated by a result of 1, while in some rare cases it will be drastically underestimated. In order to overcome this drawback, we change the test profile from the operational profile to another, suitably chosen test profile Q, that is Importance sampling technique: take a random sample $\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_N$ from the test profile Q, and evaluate ℓ using the likelihood ratios estimator

$$\widehat{\ell} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{I}_f(\mathbf{x}_i) W(\mathbf{x}_i, P, Q), \tag{4}$$

where $W(\mathbf{x}, P, Q) = f(\mathbf{x}, P)/f(\mathbf{x}, Q)$ is called the likelihood ratio. It is well known that the best way to estimate ℓ is to use the change of measure with probability distribution

$$f(\mathbf{x}, Q^*) = \frac{\mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P)}{\ell}. \tag{5}$$

Namely, by using this change of measure in (5) we have

$$\mathbf{I}_f(\mathbf{x}^{(i)}) \cdot \frac{f(\mathbf{x}^{(i)}, P)}{f(\mathbf{x}^{(i)}, Q^*)} = \ell, \tag{6}$$

for all i. In other words, the estimator (3) has zero variance, and we need to produce only $N = 1$ sample. The obvious difficulty is of course that this test profile Q^* depends on the unknown parameter ℓ, so it cannot be computed in practice. To obtain the optimal-or at least a good-change of measure from the original sampling distribution P to a new distribution Q, is the main challenge with respect to making importance sampling efficient and robust.

From equation (5), the only restriction to the probability distribution $f(\mathbf{x}, Q)$ is that $f(\mathbf{x}, Q) > 0$ for all samples $\mathbf{x}$. This is a necessary condition which serves as a guideline for choosing Q. The efficiency of the importance sampling estimate in equation (4) is dependent on an appropriate choice of the new distribution. An unfortunate choice of Q may cause the variance of equation (4) to be larger than the variance of the estimates obtained by direct simulation. For some Q, the variance may be infinite, and, hence, it is crucial to find a good Q. Since the effectiveness of importance sampling estimator totally depend on the choice of importance sampling distribution function, we often select importance sampling distribution function from a family of importance sampling distributions, and evaluate its effectiveness by computing the variance of the estimator.

The variance of $\widehat{\ell}$ in equation (4) is

$$\begin{aligned} Var(\widehat{\ell}) = \frac{1}{N}\big(&\mathbb{E}_Q[\mathbf{I}_f(\mathbf{X})W(\mathbf{X}, P, Q)]^2 \\ &-\mathbb{E}_Q^2[\mathbf{I}_f(\mathbf{X})W(\mathbf{X}, P, Q)]\big). \end{aligned}$$

We would like to minimize the variance of $\widehat{\ell}$ with respect to the optimal test profile Q^*, that is , we want to find the Q^* such that:

$$\begin{aligned} Q^* \;=\; \arg\min_Q\{\frac{1}{N}\big(&\mathbb{E}_Q[\mathbf{I}_f(\mathbf{X})W(\mathbf{X}, P, Q)]^2 \\ &-(\mathbb{E}_Q^2[\mathbf{I}_f(\mathbf{X})W(\mathbf{X}, P, Q)])\big)\} \end{aligned} \tag{7}$$

The last term in equation (7) is ℓ^2, minimizing the variance of the Importance Sampling estimator is tantamount to minimizing its first term, so equation (7) can be written as

$$\begin{aligned} Q^* &= \arg\min_Q \left\{\int \mathbf{I}_f^2(\mathbf{x}) W^2(\mathbf{x}, P, Q) f(\mathbf{x}, Q) d\mathbf{x}\right\} \\ &= \arg\min_Q \left\{\int \mathbf{I}_f^2(\mathbf{x}) W(\mathbf{x}, P, Q) \frac{f(\mathbf{x}, P)}{f(\mathbf{x}, Q)} f(\mathbf{x}, Q) d\mathbf{x}\right\} \\ &= \arg\min_Q \left\{\int \mathbf{I}_f^2(\mathbf{x}) W(\mathbf{x}, P, Q) f(\mathbf{x}, P) d\mathbf{x}\right\} \\ &= \arg\min_Q \left\{\int \mathbf{I}_f(\mathbf{x}) W(\mathbf{x}, P, Q) f(\mathbf{x}, P) d\mathbf{x}\right\} \\ &= \arg\min_Q \{\mathbb{E}_P[\mathbf{I}_f(\mathbf{X}) W(\mathbf{X}, P, Q)]\} \end{aligned} \tag{8}$$

where $\mathbf{I}_f^2(\mathbf{x}) = \mathbf{I}_f(\mathbf{x})$ since the indicator function can assume just 0 or 1, and the notation arg means that Q is the value of the test profile that minimizes the variance $Var(\widehat{\ell})$.

The expression of expectation in equation (7)

$$\mathbb{E}_P[\mathbf{I}_f(\mathbf{X})W(\mathbf{X}, P, Q)] = \int \mathbf{I}_f(\mathbf{x}) \frac{f(\mathbf{x}, P)}{f(\mathbf{x}, Q)} f(\mathbf{x}, P) d\mathbf{x} \tag{9}$$

can be seen as a measure of the Ali-Silvey distance [21] between two probability distributions $h_1(\mathbf{x})$ and $h_2(\mathbf{x})$:

$$\mathcal{D}_{AS} = \int h_1(\mathbf{x}) \frac{h_1(\mathbf{x})}{h_2(\mathbf{x})} d\mathbf{x}, \tag{10}$$

where

$$h_1(\mathbf{x}) = \frac{\mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P)}{\int \mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P) d\mathbf{x}}, \quad h_2(\mathbf{x}) = f(\mathbf{x}, Q).$$

So that the optimization problem is formulated as the minimization of the Ali-Silvey distance. In order to facilitate the calculation, we introduce a different distance measure, namely Kullback-Leibler distance [23],[24]:

$$\mathcal{D}(h_1(\mathbf{x}), h_2(\mathbf{x})) = \int h_1(\mathbf{x}) \ln \frac{h_1(\mathbf{x})}{h_2(\mathbf{x})} d\mathbf{x} \tag{11}$$

The optimal test profile problem, previously formulated as the minimization of the Ali-Silvey distance measure can be reformulated as the minimization of the Kullback-Leibler distance measure:

$$
\begin{aligned}
Q^* &= \arg\min_Q \mathcal{D}_{AS} \approx \arg\min_Q\{\mathcal{D}(h_1(\mathbf{x}), h_2(\mathbf{x}))\} \\
&= \arg\min_Q \left\{ \int \frac{\mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P)}{\int \mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P) d\mathbf{x}} \cdot \right. \\
&\qquad \left. \ln \frac{\mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P)}{f(\mathbf{x}, Q) \int \mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P) d\mathbf{x}} d\mathbf{x} \right\} \\
&= \arg\min_Q \left\{ - \int \mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P) \ln f(\mathbf{x}, Q) d\mathbf{x} \right\} \\
&= \arg\max_Q \left\{ \int \mathbf{I}_f(\mathbf{x}) f(\mathbf{x}, P) \ln f(\mathbf{x}, Q) d\mathbf{x} \right\} \\
&= \arg\max_Q \{\mathbb{E}_P[\mathbf{I}_f(\mathbf{X}) \ln f(\mathbf{X}, Q)]\} \qquad (12)
\end{aligned}
$$

By equation (3), The total probability of the test case $\mathbf{x}$ with test profile Q is

$$
\begin{aligned}
f(\mathbf{x}, Q) &= \prod_i q(x_i, x_{i+1}), \\
\text{such that} \quad & \sum_k q(x_i, k) = 1,
\end{aligned}
$$

where k runs over those states that can be reached in one step from state x_i, since all other $q(x_i, k)$ are 0, and i runs over all arcs in the test case $\mathbf{x}$. Using Lagrange multipliers $u_1, u_2, \cdots$, we obtain the maximization problem

$$
\max_Q \left\{ \mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \sum_i \ln q(X_i, X_{i+1}) \right] + \sum_i u_i \left(\sum_k q(X_i, k) - 1 \right) \right\}. \qquad (13)
$$

To find the maximum of equation (13), set the derivative with respect to $q(l, m)$ to 0, for any two states l and m, we have

$$
\frac{\mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \sum_i \mathbf{I}_{\{X_i = l, X_{i+1} = m\}} \right]}{q(l, m)} + u_i = 0.
$$

Summing over $m = 1, 2, \cdots$, gives

$$
\mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \sum_i \mathbf{I}_{\{X_i = l\}} \right] = -u_i,
$$

thus we find the following expression for the optimal transition probability $q(l, m)$ from state l to state m:

$$
q(l, m) = \frac{\mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \sum_i \mathbf{I}_{\{X_i = l, X_{i+1} = m\}} \right]}{\mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \sum_i \mathbf{I}_{\{X_i = l\}} \right]}. \qquad (14)
$$

The optimal test profile Q^* can be derive directly by the relationship (14), if its analytical solution is not feasible, by its stochastic counterpart after the simulation of N test cases.

Proposition 4.1 Let $f(\mathbf{x}, P) > 0$, then the optimal testing profile Q with the elements $q(l, m)$ such that (14) that yields an unbias estimator of $R(\pi)$.

Proof: Provided that $f(\mathbf{x}, Q) > 0$ for each $\mathbf{x}$ with $f(\mathbf{x}, P) > 0$, the expression $\frac{f(\mathbf{x}, P)}{f(\mathbf{x}, Q)} \cdot \mathbf{I}_f(\mathbf{x})$ is an unbiased estimator for ℓ, since

$$
\begin{aligned}
\mathbb{E}_Q \left[\frac{f(\mathbf{x}, P)}{f(\mathbf{x}, Q)} \cdot \mathbf{I}_f(\mathbf{x}) \right] &= \sum_{\mathbf{x}} f(\mathbf{x}, Q) \cdot \frac{f(\mathbf{x}, P)}{f(\mathbf{x}, Q)} \cdot \mathbf{I}_f(\mathbf{x}) \\
&= \sum_{\mathbf{x}} f(\mathbf{x}, P) \mathbf{I}_f(\mathbf{x}) = \mathbb{E}_P[\mathbf{I}_f(\mathbf{X})] = \ell, \qquad (15)
\end{aligned}
$$

where the operator $\mathbb{E}_Q$ denotes the expectation with respect to probability distribution Q, furthermore, by (1) we can obtain the unbias estimator of $R(\pi)$.

Lemma 4.2 Let $f(x), x \in R$ be a probability density for the random variable X, the Importance Sampling estimator of $\int h(x) dx$ is given by

$$
\widetilde{g}(X) = \frac{1}{N} \sum_{i=1}^{N} \frac{h(X_i)}{f(X_i)} \quad \text{where} \quad X_i \sim f(x),
$$

if $h(x) = \alpha f(x), \forall x \in \{x : f(x) > 0\}$ where α is some constant of proportionality, then the variance of $\widetilde{g}(X)$ is zero.

Proof: Let $s(x) = h(x)/f(x)$, then $\widetilde{g}(X) = \frac{1}{N} \sum_{i=1}^{N} s(X_i)$, the variance of $\widetilde{g}(X)$ is given by

$$
\begin{aligned}
Var\left(\widetilde{g}(X)\right) &= Var\left(\frac{1}{N} \sum_{i=1}^{N} s(X_i) \right) = \frac{Var(s(X_i))}{N} \\
&= \frac{1}{N} \int \left[s(x) - E(s(X)) \right]^2 f(x) dx,
\end{aligned}
$$

since $h(x) = \alpha f(x)$, then clearly we have $h(x)/f(x) = \alpha, \forall x \in \{x : f(x) > 0\}$, so $E[h(X)/f(X)] = \alpha$, and hence the variance of $\widetilde{g}(X)$ is zero.

Proposition 4.3 Let $f(\mathbf{X}, P) > 0$, then the optimal testing profile Q with the elements $q(l, m)$ such that (14) that yields an estimator of $R(\pi)$ with zero-variance.

Proof: By Lemma 4.2, if the likelihood ratio of the estimator of $R(\pi)$ is some constant, then we can obtain the variance of the estimator is zero.

In fact, the denominator of (14) can be rewritten as follows:

$$
\begin{aligned}
\mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \sum_i \mathbf{I}_{\{X_i = l\}} \right] &= \sum_{i=1}^{\infty} \mathbb{E}_P \left[\mathbf{I}_f(\mathbf{X}) \mathbf{I}_{\{X_i = l\}} \right] \\
&= \sum_{i=1}^{\infty} \mathbb{P}\{\mathbf{I}_f(\mathbf{X}) = 1, X_i = l\} \\
&= \sum_{i=1}^{\infty} \mathbb{P}\{\mathbf{I}_f(\mathbf{X}) = 1 | X_i = l\} \mathbb{P}\{X_i = l\} \\
&= \sum_{i=1}^{\infty} r_l \mathbb{P}\{X_i = l\}, \qquad (16)
\end{aligned}
$$

where r_l is defined as the probability of reaching the failure state (rare event) before an absorbing is reached, starting from state l. Rewrite the numerator of (14) similarly:

$$\mathbb{E}_P\left[\mathbf{I}_f(\mathbf{X})\sum_i \mathbf{I}_{\{X_i=l,X_{i+1}=m\}}\right]$$
$$= \sum_{i=1}^{\infty}\mathbb{E}_P\left[\mathbf{I}_f(\mathbf{X})\mathbf{I}_{\{X_i=l,X_{i+1}=m\}}\right]$$
$$= \sum_{i=1}^{\infty}\mathbb{P}\{\mathbf{I}_f(\mathbf{X})=1, X_i=l, X_{i+1}=m\}$$
$$= \sum_{i=1}^{\infty}\mathbb{P}\{\mathbf{I}_f(\mathbf{X})=1|X_i=l, X_{i+1}=m\}\cdot \mathbb{P}\{X_{i+1}=m|X_i=l\}\cdot\mathbb{P}\{X_i=l\}$$
$$= \sum_{i=1}^{\infty} r_m\cdot p(l,m)\cdot\mathbb{P}\{X_i=l\}, \tag{17}$$

By substituting (16) and (17) into (14), we find the following expression for the optimal transition probabilities:

$$q(l,m) = \frac{\sum_{i=1}^{\infty} r_m\cdot p(l,m)\cdot\mathbb{P}\{X_i=l\}}{\sum_{i=1}^{\infty} r_l\mathbb{P}\{X_i=l\}} = \frac{r_m\cdot p(l,m)\cdot\sum_{i=1}^{\infty}\mathbb{P}\{X_i=l\}}{r_l\cdot\sum_{i=1}^{\infty}\mathbb{P}\{X_i=l\}} = \frac{r_m p(l,m)}{r_l}. \tag{18}$$

Suppose a test case $\mathbf{x}=(x_1,x_2,\cdots,x_L,x_{L+1})$ leading to the failure state (rare event) which containing L arcs. The optimal testing profile of this test case is

$$f(\mathbf{x},Q^*)=\prod_{i=1}^{L}q(x_i,x_{i+1})$$
$$=\frac{r_{x_2}p(x_1,x_2)}{r_{x_1}}\cdot\frac{r_{x_3}p(x_2,x_3)}{r_{x_2}}\cdots\frac{r_{x_{L+1}}p(x_L,x_{L+1})}{r_L}$$
$$=\frac{r_{x_{L+1}}}{r_1}\prod_{i=1}^{L}p(x_i,x_{i+1}). \tag{19}$$

The total probability of the test case $\mathbf{x}$ with operational profile P is $f(\mathbf{x},P)=\prod_{i=1}^{L}p(x_i,x_{i+1})$. Consequently, the likelihood ratio is

$$W(\mathbf{x};P,Q^*)=\frac{f(\mathbf{x},P)}{f(\mathbf{x},Q^*)}=\frac{r_{x_1}}{r_{x_{L+1}}}. \tag{20}$$

Note that x_1 is the initial state, the last state x_{L+1} is the failure state, since $\mathbf{x}$ was defined as a test case leading to the failure state (rare event), therefore $r_{x_{L+1}}=1$. So, the likelihood ratio of the test case $\mathbf{x}$ is r_{x_1}, which is the probability of reaching the failure state (rare event) before an absorbing is reached, starting from state x_1, and thus is a constant independent of the test case. Since all sample paths (test cases) in (20) reach the failure state ($\mathbf{I}_f(\mathbf{x})=1$), and thus have same likelihood ratio r_{x_1}, by Lemma 4.2, the variance of the estimator ℓ is zero, furthermore, we have the estimator $R(\pi)$ with zero-variance.

B. A Learning Algorithm of Reliability Estimation

Note that the expectations in the right-hand side of (14) are generally not known. Using again importance sampling, we can obtain them as follows:

$$q(l,m)=\frac{\mathbb{E}_{Q_j}\left[\mathbf{I}_f(\mathbf{X})W(\mathbf{X};P,Q_j)\sum_i\mathbf{I}_{\{X_i=l,X_{i+1}=m\}}\right]}{\mathbb{E}_{Q_j}\left[\mathbf{I}_f(\mathbf{X})W(\mathbf{X};P,Q_j)\sum_i\mathbf{I}_{\{X_i=l\}}\right]},$$

where Q_j is any parameter vector, we will later interpret it as the test profile used during the jth iteration of an iterative procedure.

This optimal parameter vector Q^* (denoted Q_{j+1}) could be estimated by

$$\widehat{q}(l,m)=\frac{\sum_{m=1}^{N}\mathbf{I}_f(\mathbf{x}^{(m)})W(\mathbf{x}^{(m)};P,Q_j)\sum_i\mathbf{I}_{\{X_i=l,X_{i+1}=m\}}}{\sum_{m=1}^{N}\mathbf{I}_f(\mathbf{x}^{(m)})W(\mathbf{x}^{(m)};P,Q_j)\sum_i\mathbf{I}_{\{X_i=l\}}} \tag{21}$$

where $W(\mathbf{x};P,Q_j)=f(\mathbf{x},P)/f(\mathbf{x};Q_j)$ is the likelihood ratio, at $\mathbf{x}$ between $f(\mathbf{x},P)$ and $f(\cdot;Q_j)$. Note that the factor $\mathbf{I}_{\{X_i=l\}}$ in the denominator is just the number of visits to state l during the test case $\mathbf{X}$ runs, and that $\mathbf{I}_{\{X_i=l,X_{i+1}=m\}}$ in the numerator is the number of those visits in which the transition to state m was chosen next. Consequently, the right-hand side of (21) can be interpreted as the observed conditional probability of the transition from state l to state m.

Now we propose a learning algorithm for finding the optimal testing profile Q^*. Roughly speaking, the idea of the algorithm is the following: First, we set Q equal to operational profile P. A sample of N independent paths according to P is drawn, update the testing profile Q with (21). This is done iteratively: For $j=1,2,\cdots$ and for each line i of the testing profile Q, its current elements $q(i,j)$ are modified. In this way, a sequence of new testing profile Q is obtained, which are successively improving approximations to the optimal testing profile.

More formally, a learning algorithm of reliability estimation can be described as follows:

Algorithm 4.1 A learning algorithm of reliability estimation

1) Set $t=1$ and the initial parameter vector $\hat{Q}_0=P$.
2) Generate a sample of test case $\mathbf{X}^{(1)},\mathbf{X}^{(2)},\cdots,\mathbf{X}^{(N)}$ according to the *p*df $f(\cdot;\widehat{Q}_{t-1})$.
3) Find the new testing profile $\widetilde{Q}_t$ from (21), and using the following smoothed version adaptive updating of Q_t :
$$\widehat{Q}_t=\alpha\widetilde{Q}_t+(1-\alpha)\widehat{Q}_{t-1},\quad \alpha\in(0.4,0.9). \tag{22}$$
4) If for some $t\geq d$, say $d=4,\hat{\mathbf{v}}_t=\widehat{Q}_{t-1}=\cdots=\widehat{Q}_{t-d}$ then goto step 5; otherwise set $t=t+1$ and reiterate from step 2.

5) Let τ be the final iteration. Generate a sample $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \cdots, \mathbf{x}^{(N_1)}$ according to the $p$df $f(\cdot; \widehat{Q}_\tau)$ and estimate $R(\pi)$ via the IS estimate

$$\widehat{R}(\pi) = 1 - \frac{1}{N_1} \sum_{i=1}^{N_1} \mathbf{I}_f(\mathbf{x}^{(i)}) W(\mathbf{x}^{(i)}; P, Q_\tau). \tag{23}$$

In algorithm 4.1, the parameters N and N_1 need to be supplied in advance. Note that in steps 2-4 the optimal IS parameter is estimated. In the final step (step 5) this parameter vector is used to estimate the software reliability of interest.

Remark: Instead of updating the parameter vector $\widehat{Q}_{t-1}$ to $\widehat{Q}_t$ directly via (21), we use the smoothed version adaptive updating procedure in (22). The main reason why this smoothed updating procedure performs better is that it prevents the occurrences of 0s and 1s in the parameter vectors; once such an entry is 0 or 1, it often will remain so forever, which is undesirable. The empirical results show that a value of α between $0.4 \leq \alpha \leq 0.9$ gives the best results.

V. SIMULATION RESULTS

In order to evaluate the effectiveness of the optimal testing profile method presented in Section 4. We do simulation and compare the learning strategy based Importance sampling method with Standard method and Simulated Annealing method [18]. In standard method, the operation profile P is used to randomly generate test case. In Simulated Annealing method, the test case generated by Importance sampling method with Simulated Annealing method. In simulation we assume the parameters of concern such as the operational profile P and the failure probability $f(i,j)$ of each arc (i,j) are known. In practice they are unknown and need to be estimated in the course of software testing. As an illustration for the application of our approach, we re-investigate a Markov usage model [14]-[17] described in Fig. 1.

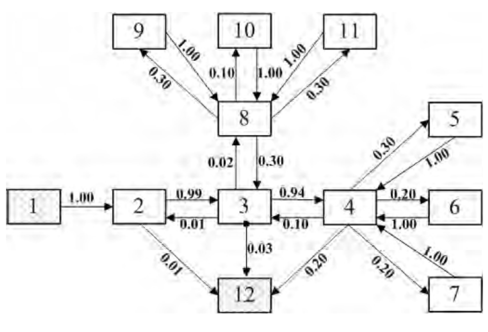

Fig. 1. A Markov Usage Model from Gutjahr[14]-[17].

The model reflects part of a train schedule program which contains 12 nodes. It has been assumed that node 6, 9 and 10 are critical operation. Each execution of the operation corresponding to arc (i, j) has a prior probability $f(i,j) > 0$ of failing, the values $f(i,j)$ have been estimated and contained in the Table 1.

We have applied our approach presented in this paper with the parameters $N = 10,000$, $\rho = 0.002$, $\alpha = 0.4$, $d = 4$. The results produced by our approach and those obtained by standard method and simulated annealing method compared to 200 test simulations. In each simulation, we have generated a sample of $N_1 = 1,000$ test cases to estimate the reliability of software. Figure 2 and figure 3 show the results of 200 test simulations obtained by three different test methods. Figure 4 show the boxplot of the results of 200 test simulations.

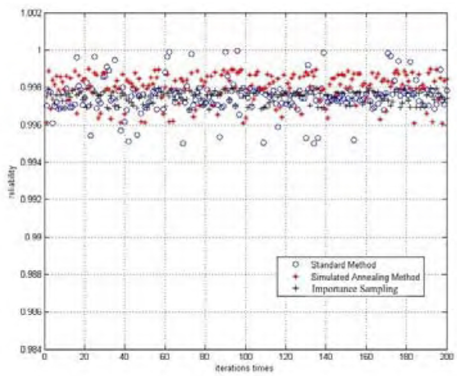

Fig. 2. The reliability estimation with three different methods.

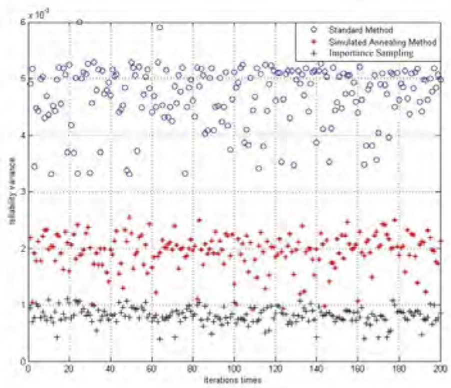

Fig. 3. The variance of estimation with three different methods.

According to the 200 test simulations, we have obtained the average value of the reliability estimation $\widehat{R}(\pi)$ by our approach is 0.9985, the average value of variance 8.32×10^{-4}, while the average value of reliability estimation according to the standard method is 0.9968 with variance 5.21×10^{-3}, and the Simulation annealing method is 0.9976 with variance 1.683×10^{-3}. In fact, zero variance by algorithm 4.1 is not

TABLE I

THE PRIOR PROBABILITY $f(i,j)$ OF FAILING

	1	2	3	4	5	6	7	8	9	10	11	12
1	0	.001	0	0	0	0	0	0	0	0	0	0
2	0	0	.01	0	0	0	0	0	0	0	0	.001
3	0	.001	0	.01	0	0	0	.001	0	0	0	.001
4	0	0	.001	0	.01	.01	.001	0	0	0	0	.001
5	0	0	0	.001	0	0	0	0	0	0	0	0
6	0	0	0	.001	0	0	0	0	0	0	0	0
7	0	0	0	.001	0	0	0	0	0	0	0	0
8	0	0	.001	0	0	0	0	0	.001	.1	.001	0
9	0	0	0	0	0	0	0	.001	0	0	0	0
10	0	0	0	0	0	0	0	.001	0	0	0	0
11	0	0	0	0	0	0	0	.001	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0

reached, since the new transition probabilities are estimated from simulation results, which obviously have a statistical error. The error can be reduced by increasing the size of sample used in algorithm 4.1.

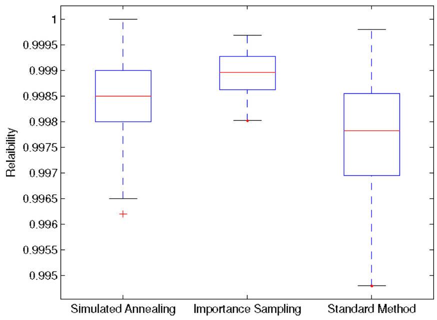

Fig. 4. Boxplot of reliability estimation with three different methods.

In table 2, we obtain the average execution number of the each critical transition with three different test method.

TABLE II

SIMULATION RESULTS FOR 200 TEST SAMPLING

	The average execution number		
	$(4,6)$	$(8,9)$	$(8,10)$
Standard Method	92.3	34.53	121.7
Simulated Annealing	164.2	145.5	253.3
Importance Sampling	289.8	101.4	402.2

It can be observed that the intended effect of favoring the test of critical operations has been achieved. The generation of test case according the operational profile P gives the critical operation 6, 10 only a small probability of being tested. Within the optimized test profile Q^* the transitions $(4,6),(8,9)$ and $(8,10)$ obtain a fair chance.

VI. CONCLUSIONS

We have developed a novel technique for estimating software reliability in a Markov software usage model, taking account of pre-information on failure probabilities. Specifically the procedure uses Importance Sampling technique based on Ali-Silvey distance. The reliability estimator with maximum precision is obtained from the test results. To show the method's performance, we have applied it to estimate the software reliability of Gutjahr's Markov usage model, the good performance of the method has been demonstrated experimentally.

However, the statistical testing models are modeled by Markov usage models, and the number of nodes is not too large to avoid state space explosion. This indicates two obvious directions for future work: extension to non-Markov state models, and developing more efficient methods for handling large state spaces such as parallel technique. Furthermore, it may be possible to provide more solid mathematical foundations, such as a proof of the convergence of the parameter vector sequence.

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